

1. Two fair dice are rolled and the absolute value of the difference of the outcomes is denoted by  $X$ . What are the possible values of  $X$ , and the probabilities associated with them?

| 2 <sup>nd</sup> \ 1 <sup>st</sup> | 1 | 2 | 3 | 4 | 5 | 6 |               |
|-----------------------------------|---|---|---|---|---|---|---------------|
| 1                                 | 0 | 1 | 2 | 3 | 4 | 5 | diff (0) = 6  |
| 2                                 | 1 | 0 | 1 | 2 | 3 | 4 | diff (1) = 10 |
| 3                                 | 2 | 1 | 0 | 1 | 2 | 3 | diff (2) = 8  |
| 4                                 | 3 | 2 | 1 | 0 | 1 | 2 | diff (3) = 6  |
| 5                                 | 4 | 3 | 2 | 1 | 0 | 1 | diff (4) = 4  |
| 6                                 | 5 | 4 | 3 | 2 | 1 | 0 | diff (5) = 2  |

ANS: (1)  $X \in \{0, 1, 2, 3, 4, 5\}$

$$(2) P(X=0) = \frac{1}{6}, P(X=1) = \frac{5}{18}, P(X=2) = \frac{2}{9}$$

$$P(X=3) = \frac{1}{6}, P(X=4) = \frac{1}{9}, P(X=5) = \frac{1}{18}$$

2.  $F$ , the distribution function of a random variable  $X$ , is given by

$$F(t) = \begin{cases} 0 & t < -1 \\ (1/4)t + 1/4 & -1 \leq t < 0 \\ 1/2 & 0 \leq t < 1 \\ (1/12)t + 7/12 & 1 \leq t < 2 \\ 1 & t \geq 2. \end{cases}$$

Then, calculate the following quantities:  $P(X < 1)$ ,  $P(X = 1)$ ,  $P(0 \leq X < 1)$ ,  $P(X > 1/2)$ ,  $P(X = 3/2)$ , and  $P(1 < X \leq 6)$ .

ANS: (1)  $P(X < 1) = \frac{1}{2}$

(2)  $P(X = 1) = \frac{1}{12} + \frac{7}{12} - \frac{1}{2} = \frac{1}{6}$

(3)  $P(0 \leq X < 1) = \frac{1}{2} - (\frac{1}{4} \times 0 + \frac{1}{4}) = \frac{1}{4}$

(4)  $P(X > \frac{1}{2}) = 1 - (\frac{1}{2}) = \frac{1}{2}$

(5)  $P(X = \frac{2}{3}) = 0$

(6)  $P(1 < X \leq 6) = 1 - (\frac{1}{12} + \frac{7}{12}) = \frac{1}{3}$

3. Assume that 12% of people have traveled internationally. A company wants to hire a new employee who has traveled internationally. How many applicants do they need to interview to have a 60% chance that at least one of the applicants has traveled internationally??

Let  $A(x)$  be at least 1 people travel internationally among  $x$  people

$$A^c(x) = (88\%)^x \quad A(x) = 1 - (88\%)^x$$

$$A(x) = 1 - (88\%)^x \geq 60\% \quad x \geq 7.167850$$

A: 8

4. In the experiment of rolling a balanced die twice, let  $X$  be the minimum of the two numbers obtained. Determine the probability mass function and the distribution function of  $X$ .

| 1 <sup>st</sup> \ 2 <sup>nd</sup> | 1 | 2 | 3 | 4 | 5 | 6 |
|-----------------------------------|---|---|---|---|---|---|
| 1                                 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2                                 | 1 | 2 | 2 | 2 | 2 | 2 |
| 3                                 | 1 | 2 | 3 | 3 | 3 | 3 |
| 4                                 | 1 | 2 | 3 | 4 | 4 | 4 |
| 5                                 | 1 | 2 | 3 | 4 | 5 | 5 |
| 6                                 | 1 | 2 | 3 | 4 | 5 | 6 |

$$P(X=1) = \frac{11}{36} \quad P(X=4) = \frac{5}{36}$$

$$P(X=2) = \frac{9}{36} \quad P(X=5) = \frac{3}{36}$$

$$P(X=3) = \frac{7}{36} \quad P(X=6) = \frac{1}{36}$$

ANS:

|                           |                 |       |                       |                 |                |
|---------------------------|-----------------|-------|-----------------------|-----------------|----------------|
| Probability mass function | $\frac{11}{36}$ | $X=1$ | Distribution function | $\frac{11}{36}$ | $X < 1$        |
|                           | $\frac{9}{36}$  | $X=2$ |                       | $\frac{20}{36}$ | $1 \leq X < 2$ |
|                           | $\frac{7}{36}$  | $X=3$ |                       | $\frac{27}{36}$ | $2 \leq X < 3$ |
|                           | $\frac{5}{36}$  | $X=4$ |                       | $\frac{32}{36}$ | $3 \leq X < 4$ |
|                           | $\frac{3}{36}$  | $X=5$ |                       | $\frac{35}{36}$ | $4 \leq X < 5$ |
|                           | $\frac{1}{36}$  | $X=6$ |                       | 1               | $5 \leq X$     |

5. In a lottery every week, 2,000,000 tickets are sold for \$1 apiece. If 4000 of these tickets pay off \$30 each, 500 pay off \$800 each, one ticket pays off \$1,200,000, What is the expected value of the winning amount for a player with a single ticket?

$$P(\text{Win } -1\$) = \frac{1995499}{2,000,000} \quad P(\text{Win } 29\$) = \frac{4000}{2,000,000}$$

$$P(\text{Win } 799\$) = \frac{500}{2,000,000} \quad P(\text{Win } 1,199,999\$) = \frac{1}{2,000,000}$$

$$E(X) = -1 \times \frac{1995499}{2,000,000} + 29 \times \frac{4000}{2,000,000} + 799 \times \frac{500}{2,000,000} + 1199999 \times \frac{1}{2,000,000}$$

$$= \frac{-280000}{2000000} = -0.14$$

ANS: -0.14

6. If  $X$  is a random number selected from the first 10 positive integers, what is  $E[X(11-X)]$ ?

$$E(11X - X^2) = 11E(X) - E(X^2)$$

$$E(X) = (1+2+3+\dots+9+10) \times \frac{1}{10} = \frac{55}{10}$$

$$E(X^2) = (1^2+2^2+3^2+\dots+9^2+10^2) \times \frac{1}{10} = \frac{385}{10}$$

$$11 \times \frac{55}{10} - \frac{385}{10} = \frac{220}{10} = 22$$

A: 22

7. Mr. Jones is about to purchase a business. There are two businesses available. The first has a daily expected profit of \$150 with standard deviation \$30, and the second has a daily expected profit of \$150 with standard deviation \$55. If Mr. Jones is interested in a business with a steady income, which should he choose?

A: Business 1, because lower the standard deviation is, more steady the data are

8. Find the variance and the standard deviation of a random variable  $X$  with distribution function

$$F(x) = \begin{cases} 0 & x < -3 \\ 3/8 & -3 \leq x < 0 \\ 3/4 & 0 \leq x < 6 \\ 1 & x \geq 6. \end{cases}$$

$$P(X = -3) = \frac{3}{8} - 0 = \frac{3}{8}$$

$$P(X = 0) = \frac{3}{4} - \frac{3}{8} = \frac{3}{8}$$

$$P(X = 6) = 1 - \frac{3}{4} = \frac{2}{8}$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$E(X^2) = (-3)^2 \times \frac{3}{8} + (0)^2 \times \frac{3}{8} + 6^2 \times \frac{2}{8} = \frac{99}{8}$$

$$E(X) = (-3) \times \frac{3}{8} + (0) \times \frac{3}{8} + 6 \times \frac{2}{8} = \frac{3}{8}$$

$$\text{Var} = \frac{99}{8} - \frac{9}{64} = \frac{783}{64}$$

A:  $\frac{783}{64}$

9. Suppose that  $X$  is a discrete random variable with  $E[X]=1$  and  $E[X(X-2)]=3$ . Find  $\text{Var}(-3X+7)=?$

$$E(X(X-2)) = E(X^2 - 2X) = E(X^2) - 2E(X) = 3$$

$$E(X^2) = 3 + 2E(X) = 3 + 2 = 5$$

$$\begin{aligned}\text{Var}(-3X+7) &= 9 \text{Var}(X) = 9 \cdot (E(X^2) - (E(X))^2) \\ &= 9 \cdot (5 - 1) = 36\end{aligned}$$

Ans: 36